

RF Link Planner

Terrain-aware network planning for Mesh Rider radios. What the tool does, how far we can trust it, and what it has already found.

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Hosted on our own Cloudflare account

14 radios

Full per-MCS data, including v4 and RM-1300/1400

90 antennas

Every row audited against manufacturer datasheets

Exact

Reproduces the official Doodle Labs estimator

In one paragraph

We have an in-house planning tool that puts Mesh Rider links on a real map. You place radios, pick the actual radio, band, antenna and mast height, and it tells you what the link will really do - throughput, margin, coverage footprint, and where it breaks. Underneath it reproduces the official Doodle Labs range estimator exactly, then adds the terrain that a distance-only calculator cannot see. It is internal-only today, and it is already finding real faults in customer logs and matching field results from Ukraine.

What you can do with it

- **Plan a link.** Place radios and get distance, best usable MCS, throughput and margin. Click a link for the terrain cross-section with the Fresnel zone drawn, so you can see which ridge is costing you.
- **Coverage maps, two ways.** **Terrain-following** for UGVs, vehicles and handhelds - what can something driving around here reach? And **flight level** for UAS, where you drag an altitude slider and coverage repaints live. It also maps the minimum altitude a drone needs to connect, point by point.
- **Mesh overlap and redundancy.** Combines every radio footprint and colours by how many radios cover each spot - one, two (survives any single outage), or three or more. Reports the area of each in square kilometres.
- **Mission routes.** Paste a waypoint list or click it on the map, set up the vehicle radio, and get percentage of route with a link, percentage at your required data rate, longest dropout in metres, handover count, and which radio serves which stretch.
- **Plan Advisor.** Describe the mission - "ground station to UAV, 10 km, 5 Mbps" - and it searches every radio, band, channel width and antenna combination, then hands back ranked builds with aiming, heights and cabling. It will also extend an existing network and site a relay when one hop cannot reach.
- **Locate by signal.** Nano and Mini radios carry no GPS. Use the OEM and Wearable units, which do, as anchors: enter the levels they report and the tool works out position from the link budget, with an honest confidence circle rather than a false pin.
- **Customer-ready reports.** One click exports a document with the map, equipment table, per-link terrain analysis and minimum UAV altitudes. Prints straight to PDF.
- **Fade margin you control.** Set 10 dB for our standard plan, 12-14 dB for field practice, higher for contested RF. Every calculation follows it and the report states the value used, so nothing goes out based on a looser number

than you quoted.

Why the numbers can be trusted

The link budget **reproduces the official Doodle Labs Range Estimation Tool exactly** - same conservative sensitivities, same per-MCS transmit powers - verified to the metre across roughly thirty test cases. On top of that it adds what a distance-only calculator cannot: real elevation data, earth curvature, Fresnel-zone clearance, diffraction, antenna radiation patterns and frequency-dependent cable loss.

The antenna audit

Our recommended-antenna spreadsheet was checked row by row against manufacturer datasheets. That work is now baked into the tool:

Result	Count	What it means
Confirmed	31	Sheet matched the manufacturer datasheet
Specs corrected	19	Real product, wrong numbers on our sheet
Not a real product	20	Family labels, platforms, or part numbers that do not exist
Unverifiable	6	No authoritative source found anywhere

Four catches worth knowing about

Most of the L-com HG4958-xx part numbers do not exist as written. Both Poynting entries are cellular antennas we had listed as Wi-Fi. The Triad BDA is rated 25 W, not the 16 W on our sheet. And one recommended sector antenna, rated 5 W maximum input, was paired with a 20 W amplifier that would have destroyed it.

Tested against reality

Ukraine field test

A colleague reported losing his link at 42-44 km: 24 dBi dish at 2 m, 3 dBi omni at 1000 m, 5 MHz channel, clear air. Modelled at his own 12-14 dB fade margin, the tool puts the cliff inside that exact window, with no adjustment:

Fade margin	Tool prediction at 42.6 km
10 dB	MCS 10, 7.4 Mbps, +10.5 dB margin
12 dB	MCS 9, 5.1 Mbps, +12.5 dB margin
14 dB	MCS 1, 2.7 Mbps, +15.5 dB margin
20 dB	No link

It also gave him two things to act on. His frequency sweep up to 2500 MHz costs about 3.5 dB, so staying near 1675 MHz is worth roughly 1.5 times the range. And the tightest point on the whole 43 km path is 200 m in front of his ground station, where a 2 m antenna only just clears - lifting that dish is the cheapest margin he can buy.

What it found in customer logs

We built parsers for Mesh Rider support bundles and ran six of them from three customer systems, about 28,000 telemetry samples:

System	Finding
Flight 7 ground station	One receive chain 10 dB down (chains at -73 and -63 dBm). The link stayed associated the whole time yet was unusable: 80% packet loss, never got past MCS 6, because a dead chain means the second data stream never works. Signal strength was never the problem.
Flight 7 channel	The 918 MHz channel was busy 70% of the time with noise peaks to -75 dBm, against 9% on a healthy reference pair.
Kratos fleet	The same 5-7 dB chain imbalance on both the ground station and the relay, against every peer - which localises it to their own antenna paths rather than the air.
Kratos relay	Linked only 69.6% of the time while hopping across three frequencies and two channel widths, against a ground station that never left one channel.

It also caught its own false alarms

A "transmit power is only 15 dBm" that turned out to be the 5 GHz Wi-Fi hotspot interface, not the mesh radio (which runs 30-32 dBm as configured). A "98.9% error rate" that was simply spectral-scan samples being counted as errors. And one of our own conclusions withdrawn after a timing audit showed two logs were captured 32 minutes apart rather than simultaneously. Being able to rule things out matters as much as finding them.

Where it is honest about its limits

- Terrain data is about 30 m resolution, so **buildings and trees are not modelled**. Urban and forest predictions run optimistic - budget extra margin there.
- Interference and rain are covered only by the fade margin, not modelled explicitly.
- Predictions are planning-grade engineering estimates, not a warranty. Field-verify critical links.
- The core link budget is thoroughly validated. The newer features - coverage modes, routes, signal-based ranging - are self-audited but have **not had independent review** yet.
- Radios without GPS can only be positioned when several of them measure each other **at the same time**. Sequential captures cannot be combined, and the tool refuses rather than guessing.

What would help most

Have a play - a 60-second tour runs on your first visit, and Help has tutorials and an FAQ. Then tell us where it is wrong, especially anywhere it disagrees with something you have actually measured: hit **Save** and send the .json with what you saw. Access control goes in before anything reaches a customer.